Department of Physics, FNSPE, CTU in Prague

# 02YELMA - Electricity and Magnetism

## Final Exam - Solutions 2

**Q1:** (a) Energy is conserved in the circuit, so the total voltage drop across the circuit elements must equal the applied voltage $V = V_R + V_L + V_C = 0$. The voltage drops are

$$V_R = RI, \quad V_L = L\frac{dI}{dt}, \quad V_C = \frac{q}{C}$$

The charge on the capacitor is related to the current by $I = dq/dt$. Thus,

$$RI + L\dot{I} + \frac{q}{C} = 0 \xrightarrow{d/dt} R\dot{I} + L\ddot{I} + \frac{1}{C}I = 0 \implies \boxed{\ddot{I} + \frac{R}{L}\dot{I} + \frac{1}{LC}I = 0}$$

(b) The governing equation describes a damped oscillation where the coefficient of the first derivative term determines the amount of damping: $2\beta = R/L$. To obtain the characteristic equation, we assume a solution of the form $I(t) = e^{\lambda t}$, which gives

$$\lambda^2 + \frac{R}{L}\lambda + \frac{1}{LC} = 0 \implies \lambda = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}.$$

For underdamped oscillations, the discriminant must be negative, which gives the condition on $R$, $L$, and $C$:

$$\boxed{R < 2\sqrt{L/C}}$$

For current to fall to half its initial value:

$$\frac{I_{\text{max}}e^{-\beta t}}{I_{\text{max}}} = \frac{1}{2} \implies e^{-\beta t} = \frac{1}{2} \implies \boxed{t = \frac{\ln 2}{\beta} = \frac{2L\ln 2}{R}}$$

(c) The total energy in the circuit is $U = U_L + U_C = \frac{1}{2}LI^2 + \frac{1}{2}CV^2$. If the capacitor is initially charged to $V_0$, then the initial energy is $U_0 = \frac{1}{2}CV_0^2$. When the current is at its maximum, the energy is stored entirely in the inductor, so $U = \frac{1}{2}LI_{\text{max}}^2$. At resonance, we know that only the resistor dissipates energy and current behaves as $I \sim I_0\sin(\omega_0 t)$. Therefore, the average energy dissopated in the resistor over one cycle is

$$P_{\text{avg}} = \frac{1}{2}RI_0^2, \quad T = \frac{2\pi}{\omega_0} \implies \Delta E = P_{\text{avg}}T = \pi RI_0^2/\omega_0.$$

Therefore, the quality factor is

$$Q = 2\pi\frac{U}{\Delta E} = 2\pi\frac{\frac{1}{2}LI_0^2}{\pi RI_0^2/\omega_0} = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} \implies \boxed{Q = \frac{1}{R}\sqrt{\frac{L}{C}}}$$

**Q2:** (a) Because of cylindrical symmetry, the magnetic field can only have a angular component, $\vec{B}(\phi, t) = B(\phi, t)\,\hat{\phi}$. Consider an Amperian loop of radius $a < s < b$ coaxial with the shells:

$$B(\phi, t)\,(2\pi s) = \mu_0 I(t) \implies \boxed{\vec{B}(\phi, t) = \frac{\mu_0 I(t)}{2\pi s}\,\hat{\phi}, \quad \text{for} \;\; a < s < b.}$$

**Q3:**

**Q4:**